

Engineering Plan: Improving Toddito/Pulse Using Koherent's Patterns

September 29, 2026

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Engineering Plan: Improving Toddito/Pulse Using Koherent's Patterns

- **Requested by:** operator, via gm (msg_67391fac_865948 + addendum msg_37d40a10_3330180, 2026-09-29)
- **Authored by:** plan (non-interactive), research dispatched to two forked subagents (Pulse/Koherent code-mining; 1987 DOS-binary variant analysis — the latter shared with the separate standalone business plan, docs/PLAN_toddito-1987-venture.md, per gm's explicit instruction to keep the two documents independent)
- **Scope:** a real, scoped engineering plan for what to build first in Pulse (repo: /Users/flybyflow/conductor/repos/pulse, live at getyourpulse.io), grounded in Koherent's patterns, Todd's actual asks, and Pulse's own current state — verified directly against the repo, not assumed from the relay.

0. Correction to the premise, checked before scoping anything

Pulse is not a lightly-touched hackathon MVP. docs/STATUS.md (snapshot 2026-07-22, confirmed current via git log — ~135 merged PRs) shows a mature, actively-developed product: a full voice-led diagnostic pipeline (ElevenLabs → Claude scoring → Supabase → report), an invite-only alpha with real accounts/auth/referrals (Track B, shipped), a full ops console (ADR 003, shipped), voice-infra extraction already shipped, and — most relevant to this plan — **a 1987-methodology fidelity pass already done:** docs/methodology/mis-glossary-distilled.md and docs/methodology/pulse-vs-1987-fidelity-memo.md (8 named findings, F1–F8, most already shipped). Any plan that treats this as green-field “mining Koherent to build the diagnostic engine” would be redundant with real, already-shipped work. This plan is scoped against what's actually still open, not what's already done.

1. Koherent V1 (`koherentai-main`) — what's real, corrected from the

operator's framing

The operator said “there's a lot of things there that are usable” — checked directly, this is **overstated for the bulk of the repo, accurate for a few specific functions**. The app shell is the **unmodified Vercel “Next.js AI Chatbot” template** (its own README still reads “Next.js AI Chatbot”; OpenAI + Vercel KV; “We Love Koherent ❤️” is the only edit to the README itself). `instructions.json` (32KB, the file most worth reading per the operator) is a single OpenAI Assistants-API config — `name: "Math Tutor - gpt3"` (never renamed from a template default) — whose 32KB `instructions` string is **pure vision/prompt text**: Koherent's pitch plus a catalog of ~50 named psychology/facilitation frameworks (Big Five, MBTI, DISC, Belbin, Hogan, FIRO-B, Johari Window, World Café, Open Space, Theory U, NVC, Sociocracy, Collective Presencing). No schema, no scoring logic — reference material, not code.

Real, reusable, working code does exist, in `lib/chat/actions.tsx` (777 lines):

- `calculateBigFiveScores` / `calculateTraitScore` (:740, :756) — averages 1–20 sub-trait answers into a 0–100 standardized score. Small, portable, generic scoring pattern.
- `formatProfileForPrompt` (:702) — builds a profile string (DISC/MBTI/ Big Five type) injected into a system prompt.
- `getMentionedUsersData` / `formatAllUserData` (:291, :306) + `extractEmails` (:269) — **the actual “profile people, tag them, converse about the space between them” mechanism**: detects @-mentioned emails in a message, pulls each person's stored profile, and formats multi-person context into the prompt. This is the one piece with a concrete, scoped reuse case for Pulse (§5c below).
- `lib/chat/ai-instructions.ts` — a slightly newer copy of the same prompt, keyed `en:` — implies a localization pattern already existed in Koherent, relevant context for Todd's multi-language (French) ask, though not itself reusable code.

2. “What / So What / Now What” — not found in V1; a genuine, separate

design opportunity, not core scope here

Full-repo grep of `koherentai-main` (`app/`, `components/`, `lib/`, `instructions.json`) for “so what” / “now what” / “nugget”: **zero hits**. Neither the prompt text nor any schema/field/component name preserves the 6-field structure (What / So What / Now What-Self / Now What-Other / Now What-Whole / Nugget). Per the operator’s own later addendum (`msg_37d40a10_3330180`), this pattern belongs to **Koherent V2** (a separate repo, not checked in this pass) — the daily “roast me / serenade me” flashcard exercises, not V1’s chat tool.

Checked against Pulse’s existing report structure for overlap — genuine gap, not redundant. Pulse already has `src/components/report/focus-of-concern.ts` (838 lines): each row is `{key, priority, why, practice}` — org-level diagnostic priority, rationale, and a concrete example, transcribed from the actual 1987 MIS tables. This answers “what needs attention in the organization and why,” at the **organization** level. What/So-What/Now-What (Self/Other/Whole/Nugget) is an **individual reflective-coaching** structure — different altitude entirely. Adopting it would be additive, not a rebuild of something that already exists under a different name — but it is real new product scope (a personal-takeaway layer on top of an org-diagnostic report), not a small addition, and deserves its own design pass before being built. **Recommended: flag as a real future feature, don’t fold into this plan’s “build first” scope.**

3. “Collective intelligence” — out of scope for this plan

Zero trace in Pulse (`src/`, `docs/` grepped) or in `koherentai-main`. Confirmed V3-only, per the operator’s addendum — the pattern lives in `relational05` (not researched in this pass; that repo is explicitly task 5’s territory, the BSHR synthesis across all three Koherent versions

- Toddito + the 1987 system, sequenced after this plan per gm’s own instruction).

4. Pulse's own 1987-fidelity work, already done — don't duplicate

`docs/methodology/pulse-vs-1987-fidelity-memo.md`'s 8 findings, checked directly:

#	Finding	Status
F1	Nurture bar pole order inverted	🔴 fixed, shipped
F2	Culture count (11 Full vs 14 Lite)	✅ shipped (14 everywhere)
F3	SCORING.md doc drift	✅ fixed (docs-only)
F4	Strategy dims (4 linguistic vs 5 bipolar)	✅ shipped
F5	Traditional culture excluded	🟢 by design, matches legacy
F6	Report-substance gaps (Growth Engine/Crisis copy, discouraged cultures, Focus of Concern)	mostly shipped; two-tier risk + numeric-vs-qualitative display still open, untracked as a numbered decision
F7	Probe audit (Lite has 0 leadership probes vs 5 scored dims)	Lite v2 (P03→P22) shipped; her manual ElevenLabs prompt update still pending
F8	THOR weight divergence	🔴 OPEN — see §5

5. The highest-value concrete item: F8, and its real connection to

today's 1987-binary research

F8, stated precisely: the ported THOR weights (originally pulled from a third-party `Mana-matrix/THOR` GitHub repo) do **not** reproduce two real legacy verdicts — case A: legacy scored MODULATION, the current engine ranks STALL first; case B: legacy scored ACCELERATION→PROJECTILE, the current engine ranks AUTO_PILOT first. Encoded as `it.fails` documented- divergence tests in `src/lib/scoring/__fixtures__/thor-score.test.ts`. `docs/STATUS.md`'s own

words: **"THOR panel = directional until weights re-derived."** This is a real correctness gap in a shipping product — every report's THOR panel is currently honest about being directional (per F8's own mitigation), but it is not yet accurate.

The memo's own lead #2 (checked directly): the legacy program ships **per-industry weight files** named `GP{letter}SCORE.ASF`, and the two ground-truth test cases came from *different* industry variants — meaning the current matrix is very likely one industry's weights applied universally, not the legacy system's actual per-variant behavior.

This is where the two forked research passes connect, not coincidentally: the dedicated 1987-binary analysis (run in parallel for the standalone business plan, `docs/PLAN_toddito-1987-venture.md`) found these exact files, confirmed present, in `/Users/flybyflow/conductor/repos/mic-program-toddito`:

- 8 real lettered variants (A, B, C, D, M, R, S, T), each a `GP{letter}QUEST/RISK/SCORE/LIB.ASF` set — `GPxSCORE.ASF` is real, not hypothesized.
- Confirmed, evidence-based structural finding: the 8 variants group into **3 underlying content-families** by shared questionnaire/report text — {A, M}, {B, S}, {C, D, R, T} — meaning the RISK/SCORE weight matrices likely differ *within* a family too, or the families themselves may be the real "industry" boundary. Not yet determined which; see the venture plan for the full evidentiary trail.
- **RESOLVED (bshr, 2026-09-29):** the letter↔industry mapping is in the binaries themselves, not a Todd-dependent unknown. `GPDDMS.ASF` maps Manufacturing/Service/Retail/Large Manufacturing/Large Service/Large Retail/Restaurant/Large Restaurant directly to GPM/GPS/GPR/GPA/GPB/GPC/GPT/GPD — i.e. **A = Large Manufacturing, B = Large Service, C = Large Retail, D = Large Restaurant, M = Manufacturing, R = Retail, S = Service, T = Restaurant**. The earlier "Todd is the real source" framing (below, pre-correction) was wrong; the CPCSP.EXE industry-selection menu it cited is a red herring from an unrelated module, superseded by this direct hit. Residual open item (not a Todd question): confirm Pulse's own org intake captures an industry field that maps onto these 8 categories — a data check, not a design blocker.

Real, scoped engineering task (recommended #1 priority, contingent on schedule — see §7 ranking):

1. Reverse-engineer the `.ASF` binary format on the smallest file first (likely `GPxLIB.ASF` or `GPxRISK.ASF` — not yet inspected at the byte- layout level in either research pass; this is genuinely new work, not yet scoped to file-format spec depth).
2. Extract the weight matrix per lettered variant / content-family.
3. Re-run `thor-score.test.ts`'s two ground-truth cases against the variant-specific weights (not the universal matrix) and check whether divergence closes.
4. If it does: this becomes a real per-industry (or per-family) weight selection in the scoring pipeline, not a single hardcoded matrix — a genuinely new capability, not just a bugfix.
5. **No longer Todd-gated:** the letter↔industry mapping is resolved (above). What's left to confirm before this fully closes: whether Pulse's respondent/org intake records an industry classification that maps onto the 8 categories above — a Pulse-data check, not a Todd-sourced unknown.

6. Security backlog — flagged prominently, not buried, because this is

where real risk lives

`docs/SECURITY.md` (snapshot 2026-07-15) lists **6 S1 findings, verified still ALL open** (checked `git log` since 2026-07-15 for any SEC-0x / prompt-injection / rate-limit / webhook-replay / fail-closed commit — zero matches): transcript prompt-injection into the scoring model (highest priority — `src/lib/scoring/prompt.ts` is on the repo's own STOP list), an unauthenticated score endpoint (cost/DoS exposure), a public config route leaking client/org PII and minting billable ElevenLabs signed URLs, webhook replay (no timestamp-freshness check), silent unsigned- webhook fallback, and no rate limiting anywhere in `src/`.

Directly relevant to this plan's priority call: the security doc's own "Full-pass checklist" explicitly still lists "**Privacy/terms (Instinct AI LLC)**" as owed before public launch. This confirms, independently, that **OD6 from the 2026-06-17 CEO review — the hard gate, one-way-door decision on biometric voice-data ToS/privacy — is still unresolved more than three months later.** Pulse captures and behaviorally analyzes voice (biometric data); Todd is in the data chain; the June review was explicit

that this must not be bypassed by momentum. This plan does not recommend any feature work that assumes or moves toward wider distribution until this is resolved — consistent with, not a new restriction beyond, what was already decided in June.

7. Application Suite — Consulting Communicator, blocked not scoped

Todd's own most-differentiated idea ("I've never seen anything like that ever" — the 16-question consultant self-diagnostic) remains blocked on OD4: Mo needs the full application doc from Todd, not yet received per any evidence checked in this pass. **One new, unconfirmed candidate connection, worth flagging not asserting:** the 1987 binaries' separate "Client Relationship Module" (`CPCSP.EXE` / `GPDDMS.AWS` , confirmed by its own marketing text — "a diagnosis of the extent to which you have managed to build trust and credibility with key decision makers in the client organization") sounds structurally adjacent to the Consulting Communicator concept. **Sharper context (bshr, 2026-09-29):** trust/credibility is one of six diagnosed dimensions in `CPCSP.EXE` (Development Stage, Strategy, Structure, Culture, Leadership Style, Consultant-Client Relationship), not the whole module — narrows, doesn't confirm, the adjacency. Not verified as the same thing — surfaced for Todd to confirm or rule out when the OD4 conversation happens, not treated as settled.

8. Recommended scope — what to build first, ranked

1. **Resolve OD6 (privacy/terms for biometric data) + close the 6 S1 security findings.** Not exciting, explicitly flagged as a hard gate 3+ months ago, still open — the highest-consequence, lowest-visibility risk in this whole plan. Blocks nothing else on this list technically, but should not keep losing priority to feature work.
2. **F8 THOR weight reverse-engineering (§5).** Real, concrete, directly connects today's Koherent-mining research to the 1987-system research — the rare case where "mining the old system" produces a specific, testable engineering deliverable rather than just inspiration. Start the `.ASF` format reverse-engineering now; the industry-letter mapping is resolved (§5), so nothing in this item is Todd-gated anymore — the only residual check is whether Pulse's own org data records a matching industry field.

3. **Small, scoped Koherent V1 reuse: the multi-person @-mention context pattern (§1).** Worth a real look for Pulse's existing group/consultant multi-respondent reports — check whether `getMentionedUsersData`'s pattern (detect a mention, pull stored context, format into a prompt) maps cleanly onto Pulse's `engagement` / `session` model. Small, bounded, not a rewrite.
4. **What/So-What/Now-What (§2) — recommend a dedicated design review before building, not slotted into this plan's build scope.** Real opportunity, different altitude from what exists, deserves its own `/plan-design-review` pass the way the community-brain rebuild got one, not squeezed in here as an afterthought.
5. **Consulting Communicator (§7) — blocked on Todd (OD4).** Do not scope blind; flag as the next thing to unblock once his call happens.

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Mode: HOLD SCOPE. This is improvement work on an existing, mature, actively-developed product with real open findings (F8) and real open risk (the security backlog) already on record — the job here is correctly prioritizing and connecting what's already known, not inventing new ambition. Scope-expansion framing would be the wrong instinct against a backlog this concrete.

Step 0 — Premise Challenge: real. The operator's actual ask ("start improving Toddito now, not just keep researching") is legitimate — but "improve" concretely means closing F8 and the security backlog before it means porting new Koherent patterns, since those are the two items with the clearest do-nothing cost (a THOR panel that's honest about being "directional" is still wrong; unresolved S1 findings are a real, standing exposure, not a hypothetical one).

Existing leverage: substantial — `thor-score.test.ts`'s ground-truth harness already exists and is exactly the verification mechanism §5's task needs; `getMentionedUsersData`'s pattern is small and portable, not a new subsystem.

ENG REVIEW — Scope Challenge: the one real technical unknown in this plan is the `.ASF` binary format itself — neither research pass inspected it at the byte level, so §5's step 1 (format reverse-engineering) has a real, not-yet-bounded effort estimate. Recommend build's first concrete step be a short timeboxed spike (e.g. half a day) on one `.ASF` file to establish whether the format is quickly tractable (a simple fixed-width

or delimited record format) or genuinely obscure (proprietary binary encoding needing more serious reverse-engineering) — and report back before committing to the rest of §5's sequence.

Test Review: §5's fix is self-verifying by construction — `thor-score.test.ts`'s two `it.fails` cases are the pass/fail criteria already in the repo; no new test framework needed, just re-running an existing one against corrected weights.

VERDICT: CLEARED for the ranked scope in §8, items 1–3 buildable now (1 and 3 need no new research; 2's format-reverse-engineering spike can start immediately, its industry-mapping half needs Todd). Items 4–5 explicitly deferred pending, respectively, a dedicated design review and Todd's OD4 response — not silently folded into “build first” scope.

UNRESOLVED DECISIONS:

- Is the operator/Mo comfortable with security-backlog closure (§6, item 1 in §8) taking priority over new feature work, given it was already flagged as a hard gate 3+ months ago and remains open?
- `.ASF` binary format: tractable or not — needs the timeboxed spike before §5's full sequence can be estimated honestly.
- ~~Which 1987 letter/family maps to which industry~~ — **resolved** (§5, bshr 2026-09-29). Residual: confirm Pulse's org intake has a matching industry field — a data check, not an open decision.
- Does build agree with deferring What/So-What/Now-What to its own design review rather than building it now, given the operator's “start improving now” framing might read as wanting it included?